Baptiste Ferrere

Ph.D. Student in Applied Mathematics

Paris, France
E *baptiste706@gmail.com*
gs *Google Scholar*

Research Profile

I develop mathematically grounded methods for interpretable machine learning using functional decompositions, Hilbert-space methods, and nonparametric statistics. I focus on interaction effects under dependent inputs, their statistical learning theory, and dependence-aware explanations.

Publications

2026 **Generalized Functional ANOVA in Closed-Form: A Unified View of Additive Explanations**
Baptiste Ferrere, Nicolas Bousquet, Fabrice Gamboa, Jean-Michel Loubes.
arXiv:2605.18422.

2026 **Exact Functional ANOVA Decomposition for Categorical Inputs Models**
Baptiste Ferrere, Nicolas Bousquet, Fabrice Gamboa, Jean-Michel Loubes, Joseph Muré.
International Conference on Machine Learning (ICML 2026) – **Oral presentation**.

2025 **Fourier Analysis on the Boolean Hypercube via Hoeffding Functional Decomposition**
Baptiste Ferrere, Nicolas Bousquet, Fabrice Gamboa, Jean-Michel Loubes, Joseph Muré.
arXiv:2510.07088.

Doctoral Research

Dec. 2024– **Ph.D. in Applied Mathematics (CIFRE)**, *Université de Toulouse & EDF*, Toulouse & Chatou, France
Research on functional decompositions and nonparametric interaction models for explainable machine learning under dependent inputs.
Academic supervisors: Fabrice Gamboa and Jean-Michel Loubes.
Industrial supervisors: Nicolas Bousquet and Joseph Muré.

Ongoing Research

- Minimax-optimal rates for nonparametric interaction models with learned dependence structure.
- Interaction-based explanations for transformer architectures.

Education

2023–2024 **M.Sc. in Statistical Learning**, *Institut Polytechnique de Paris*, Palaiseau, France
Learning Theory, High-dimensional Statistics, Optimization

2021–2024 **Engineering Degree**, *ENSAE Paris*, Palaiseau, France
Statistics and Machine Learning.

2019–2021 **MPSI–MP Preparatory Classes**, *Lycée Sainte-Geneviève*, Versailles, France
Intensive mathematics and physics curriculum; preparation for the French grandes écoles entrance examinations.

Research Internships

2024 **Research Intern**, *EDF R&D*, Chatou, France
Generalized functional ANOVA for correlated inputs and Boolean models.

2023 **Research Intern**, *CREST*, Palaiseau, France
Restless linear bandits and exploration strategies under mixing assumptions; supervised by Azadeh Khaleghi.

Teaching

2026 **Teaching Assistant – Statistics and Machine Learning**, *CentraleSupélec*, Paris-Saclay, France

Talks & Scientific Events

Jul. 2026 **PILearnWater ANITI Chair Meeting**, Toulouse, France.
Apr. 2026 **MASCOT-NUM 2026 – Annual RT-UQ Meeting**, ENSAI, Rennes (Bruz), France.
Mar. 2026 **SIAM Conference on Uncertainty Quantification**, Minneapolis, MN, USA.
Feb. 2026 **ANITI Days 2026**, Toulouse, France.
Jun. 2025 **56th Journées de Statistique de la SFdS (JdS 2025)**, Marseille, France.
Apr. 2025 **MASCOT-NUM 2025 – Annual RT-UQ Meeting**, Grenoble, France.

Skills & Languages

Technical Python, LaTeX
Languages French (native), English (C1)